



|     |                                                                                                                                                                                                                                 |   |   |   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 11. | Monostable multivibrator has a _____ state.<br>(A) Stable (B) Unstable<br>(C) one stable and a quasi-stable (D) Independent                                                                                                     | 1 | 2 | 3 |
| 12. | _____ is a circuit with a predetermined DC level added to the output voltage<br>(A) Clamper (B) Positive clipper<br>(C) Halfwave rectifier (D) Coupler                                                                          | 1 | 1 | 3 |
| 13. | Mention the process of transferring the conductor pattern from the film master onto the copper-clad base material or any other metal-clad laminate.<br>(A) Artwork Generation (B) Image Transfer<br>(C) Testing (D) Fabrication | 1 | 2 | 4 |
| 14. | Recall the terminology of PCB which represents a thin photo-sensitive polymer that supports the photographic pattern of single traces or IC pads for etching<br>(A) Prepreg (B) Etching<br>(C) Photo-resist (D) Solder mask     | 1 | 2 | 4 |
| 15. | _____ is a thin layer covering copper traces on both the top side and bottom side to help ensure PCB reliability and high performance.<br>(A) Soldermask (B) prepreg<br>(C) base (D) Via                                        | 1 | 2 | 4 |
| 16. | _____ is based on a subtractive method that involves the removal of copper from undesirable areas to achieve the desired circuit patterns.<br>(A) Prepreg (B) Etching<br>(C) Photo-resist (D) Solder mask                       | 1 | 2 | 4 |
| 17. | Choose the device that overcomes the problems of heat distribution and heat removal in the system utilizing integrated circuits.<br>(A) Shielding (B) Insulation<br>(C) Heat sink (D) Soldering                                 | 1 | 2 | 5 |
| 18. | _____ is the interference of two neighboring signals.<br>(A) Cross-talk (B) Reflections<br>(C) Line noise (D) Ground noise                                                                                                      | 1 | 2 | 5 |
| 19. | _____ is a process in PCB design where the components are placed along with proper routing to ensure electrical connection.<br>(A) Schematic (B) Placement and routing<br>(C) mounting (D) lamination                           | 1 | 2 | 5 |
| 20. | _____ represents the printed wiring on the substrate, the physical size and location of all electronic and mechanical components<br>(A) Connection (B) Printed Circuit Layout<br>(C) Nets (D) Wiring                            | 1 | 2 | 5 |

**PART - B (5 × 8 = 40 Marks)**

Answer **all** Questions

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks | BL | CO |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|
| 21. | (a) (i) Derive the expressions for drift current and diffusion current. (5 marks)<br>(ii) Calculate the conductivity of silicon in intrinsic condition at room temperature of 300° K. Given that $n_i=1.5 \times 10^{10} \text{cm}^{-3}$ , $\mu_n=1300 \text{cm}^2/\text{V-s}$ , $\mu_p=500 \text{cm}^2/\text{V-s}$ (3 marks)<br><b>(OR)</b><br>(b) (i) With the aid of neat diagrams, discuss the operation of Enhancement MOSFET. (6 marks)<br>(ii) Differentiate BJT and MOSFET. (2 marks) | 8     | 3  | 1  |

- |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 22. | (a) Elaborate on the operation of the Gunn diode and Schottky diode with neat diagrams.<br><div style="text-align: center;"><b>(OR)</b></div> (b) Describe the construction, working, and applications of Silicon Controlled Rectifier.                                                                                                                                                                                                                                                                                                     | 8 | 2 | 2 |
| 23. | (a) With neat diagrams, explain in detail, the operation of an Astable multivibrator using transistors.<br><div style="text-align: center;"><b>(OR)</b></div> (b) (i) Two power supplies A and B are available in the market. Power supply <b>A</b> has no load and full load voltage of 30 V and 25 V respectively, whereas the values are 30V and 29V for supply <b>B</b> . Evaluate and identify the better power supply based on voltage regulation.<br>(ii) Draw and explain the working of a positive clipper with sinusoidal input . | 8 | 3 | 3 |
| 24. | (a) Explain the various steps for the fabrication of a single-sided PCB.<br><div style="text-align: center;"><b>(OR)</b></div> (b) (i) Comment on PCB Stress Test Analysis. List the various PCB Stress Test Types, issues, and potential failures. (4 marks)<br>(ii) Categorize the PCB boards based on their use and applications. (4 marks)                                                                                                                                                                                              | 8 | 2 | 4 |
| 25. | (a) (i) Give an account of the environmental factors of PCB Design. (4 marks)<br>(ii) Elaborate on the general rules available for design for Analog PCB. (4 marks)<br><div style="text-align: center;"><b>(OR)</b></div> (b) (i) Discuss the various design rules available for High-frequency circuits. (4 marks)<br>(ii) Comment on the problems occurring in the design of digital PCB. (4 marks)                                                                                                                                       | 8 | 2 | 5 |

**PART - C (1 × 15 = 15 Marks)**

Answer **any 1** Questions

**Marks BL CO**

- |     |                                                                                                                                                                        |    |   |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|---|
| 26. | Illustrate with neat diagrams, the steps involved in the monolithic IC fabrication process.                                                                            | 15 | 2 | 2 |
| 27. | (i) With the aid of neat diagrams, explain the operation of Switched Mode Power Supply (SMPS). (10 marks)<br>(ii) Explain the working of a Digital voltmeter (5 marks) | 15 | 2 | 3 |

\* \* \* \* \*